Week 11
Mobile robot dynamics

ENN581 Robot Motion, Control and Planning

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ACKNOWLEDGEMENT OF TRADITIONAL OWNERS

QUT acknowledges the Turrbal and Yugara, as the First Nations owners of the lands where QUT now stands. We pay respect to their Elders, lores, customs and creation spirits. We recognise that these lands have always been places of teaching, research and learning.

QUT acknowledges the important role Aboriginal and Torres Strait Islander people play within the QUT community.



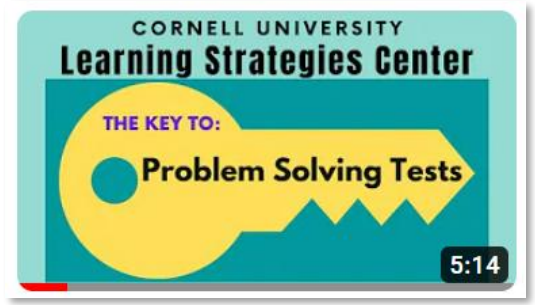
Exam Prep Tips

Getting stuck is good!

- When you get stuck, DO NOT STOP. Persist until you get to an answer (even if it's incorrect).
- Then figure out the correct solution with your notes / lecture materials / a friend.
- Watch **this video** (5 min) for why getting an answer first is so important.

Try 'Eating the Frog' to beat procrastination

- Meaning, do the hardest task of the day first.
- It might be a problem you know you'll get stuck on!
- More on eating frogs **here**.





See the full list of tips in the **Guide to Succeeding in Engineering Exams**

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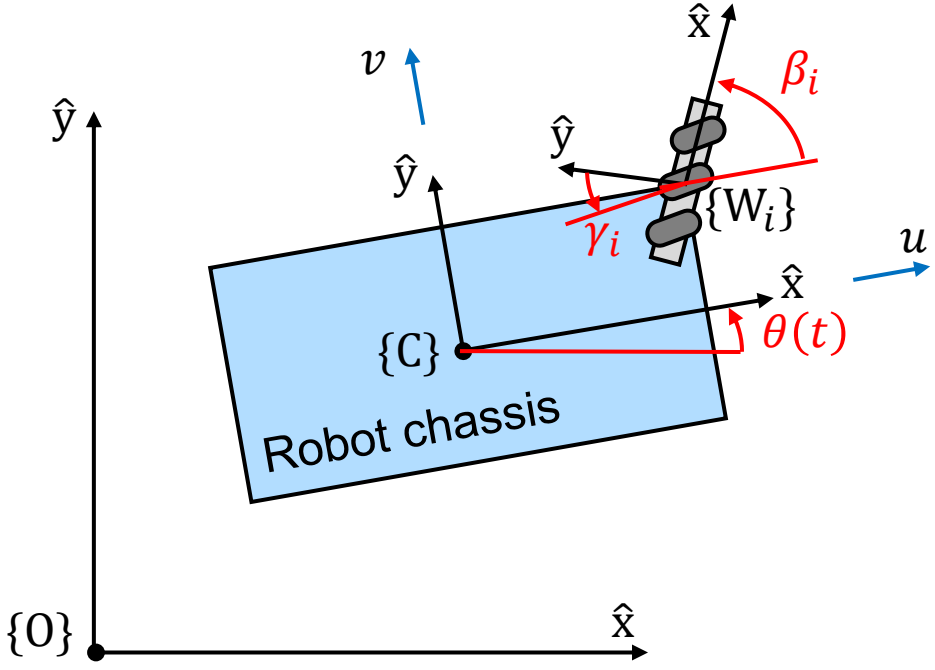
In the previous lecture, we learnt

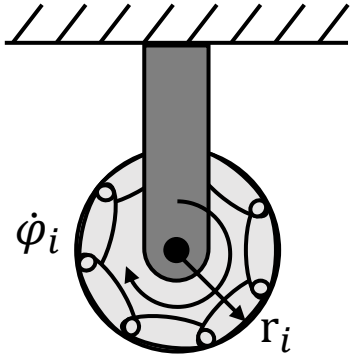
- The kinematics problem in the context of wheeled robots
- Generalised model of a wheel
- Different types of wheels and robots with wheels
- Constraints imposed by wheels



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Generalized kinematics model of a wheeled robot





We want to determine the relation between angular velocity of the wheel, $\dot{\varphi}_i$

and the desired chassis velocity, ${}^{O}v_C = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}$

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The generalised model of a wheeled robot

$$\dot{\varphi}_i = \begin{bmatrix} \frac{1}{r_i} & \frac{\tan\gamma_i}{r_i} \end{bmatrix} \begin{bmatrix} \cos\beta_i & \sin\beta_i \\ -\sin\beta_i & \cos\beta_i \end{bmatrix} \begin{bmatrix} 1 & 0 & -{}^{C}y_{W_i} \\ 0 & 1 & {}^{C}x_{W_i} \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}$$

$\dot{\varphi}_i$: the angular velocity of the wheel
r_i : the radius of the wheel
γ_i : the angle between $v_{free-roll}$ and the normal to the plane of the wheel
β_i : the angle between the v_{drive} and the robot chassis
${}^{C}x_{W_i}$ and ${}^{C}y_{W_i}$: the position of the wheel with respect to the chassis
θ : the orientation of the robot chassis with respect to a global inertial frame of reference
$\dot{x}$, $\dot{y}$, $\dot{\theta}$: the velocity of the robot chassis in the global inertial frame of reference

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Today's lecture

- Dynamics – equations of motion



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Dynamics of wheeled robots



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What is the dynamics problem?

- We are interested in obtaining a relation between the cause of motion (torques/forces on the robot) and the consequence (trajectory of the robot)

$$\mathbf{Q} = function(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, \ddot{\boldsymbol{\xi}})$$

Here $\boldsymbol{\xi}$ represents the pose of the robot in an inertial frame, $\boldsymbol{\xi} = (\mathrm{x}, \mathrm{y}, \theta)^{\mathsf{T}}$

and $\mathbf{Q}$ is the net force and torque on the robot expressed in an inertial frame

$$\mathbf{Q} = (f_{\mathrm{x}}, f_{\mathrm{y}}, \tau_z)^{\mathsf{T}}$$



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Forward Dynamics

- For a given input force-torque vector $\mathbf{Q}$, what is the trajectory $(\xi, \dot{\xi}, \ddot{\xi})$ of the robot?
- Useful for simulating or analysing the motion of the robot.



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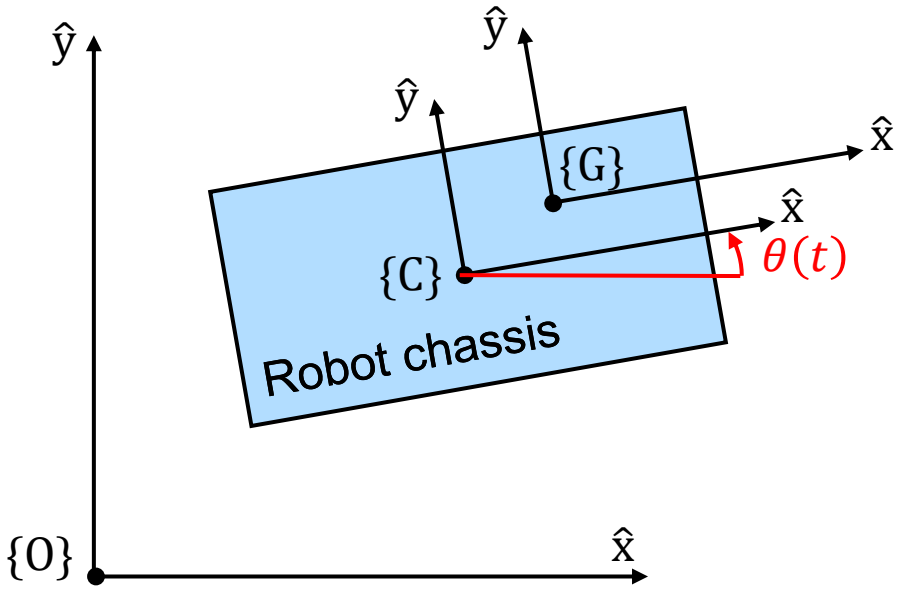
Inverse Dynamics

- For a desired trajectory $(\xi, \dot{\xi}, \ddot{\xi})$, what is the required force and torque on the robot?
- Useful for controlling the robot.



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Setup of the problem



{O} : a global inertial frame of reference
{C} : a frame attached to the robot chassis
{G} : a frame located at the centre of mass of the robot, with the same orientation as {C}
m : the mass of the robot
${}^{\{G\}}I_{zz}$: moment of inertial of the robot when rotating about an axis that is perpendicular to the ground plane and passing through the centre of mass.

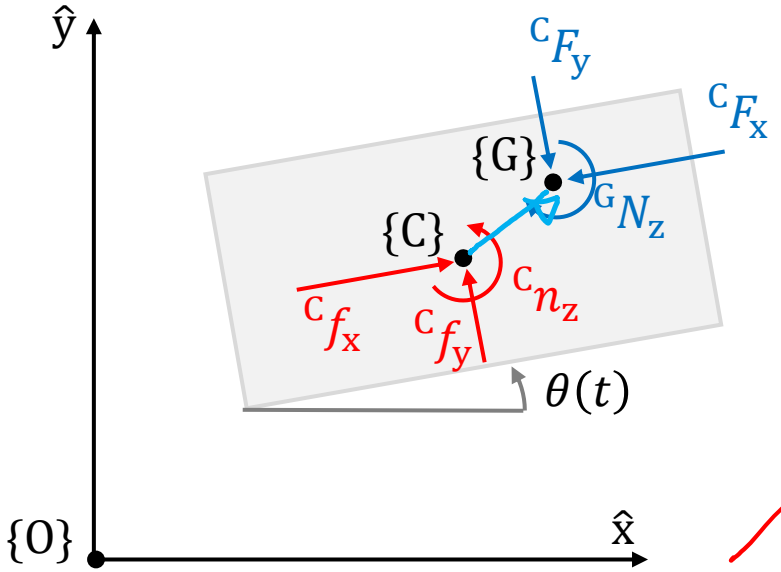
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Newton-Euler formulation



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Recall that the NE method is based on balance of forces and moments



$P_{c\,to\,G}$

$${}^{c}f_x - {}^{c}F_x = 0 \rightarrow {}^{c}F_x = {}^{c}f_x$$

$${}^{c}f_y - {}^{c}F_y = 0 \rightarrow {}^{c}F_y = {}^{c}f_y$$

$${}^{G}N_z - {}^{c}n_z - \left(-{}^{c}P_{c\,to\,G} \times {}^{c}f_c\right) = 0$$

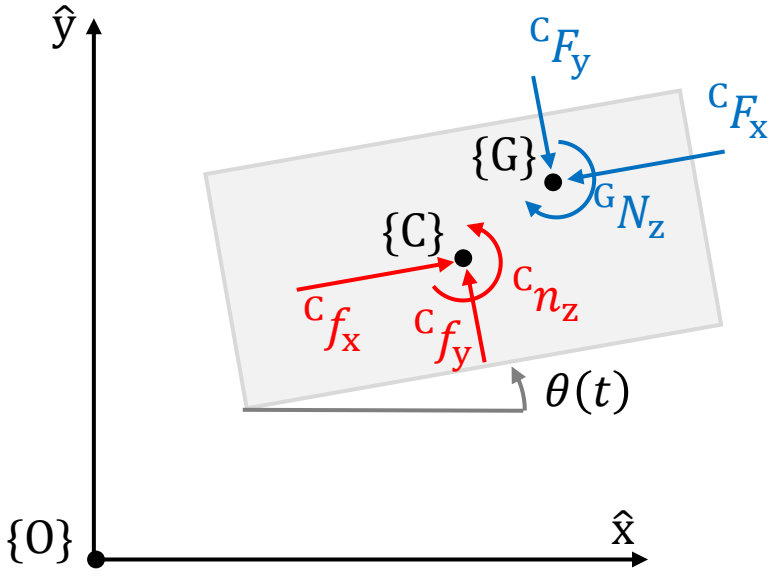
$${}^{c}f_c = \begin{Bmatrix} {}^{c}f_x \\ {}^{c}f_y \end{Bmatrix}$$

Balance moments about {G}

$${}^{G}N_z = {}^{c}n_z + \left({}^{c}P_{Gy}\,{}^{c}f_{cx} - {}^{c}P_{Gx}\,{}^{c}f_y\right)$$

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Recall that the NE method is based on balance of forces and moments



$$^{C}F_{G} - {}^{C}f_{C} = 0 \quad \begin{cases} {}^{C}F_{x} = {}^{C}f_{x} \\ {}^{C}F_{y} = {}^{C}f_{y} \end{cases}$$

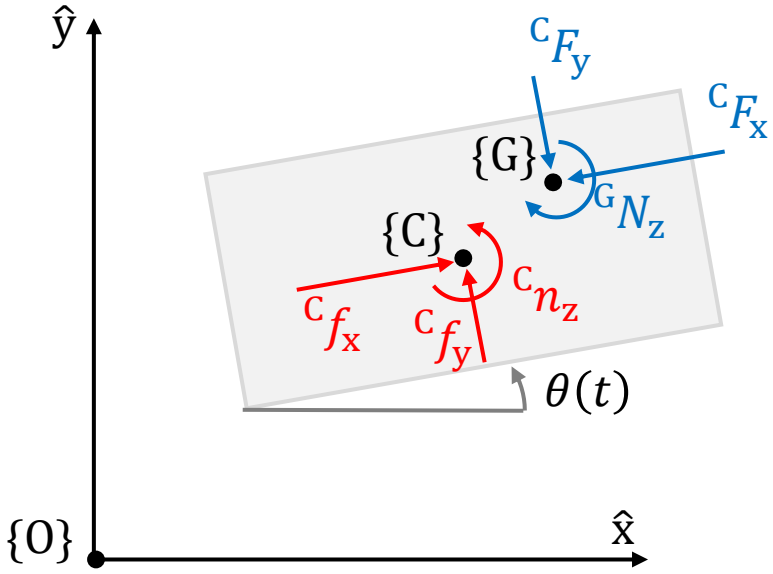
$$^{C}N_{G} - {}^{C}n_{C} - (-{}^{C}P_{CG}) \times {}^{C}f_{C} = 0 \Rightarrow {}^{C}N_{z} = {}^{C}n_{z} + ({}^{C}y_{CG}\,{}^{C}f_{x} - {}^{C}x_{CG}\,{}^{C}f_{y})$$

Note: All the terms here are in the chassis frame {C}.

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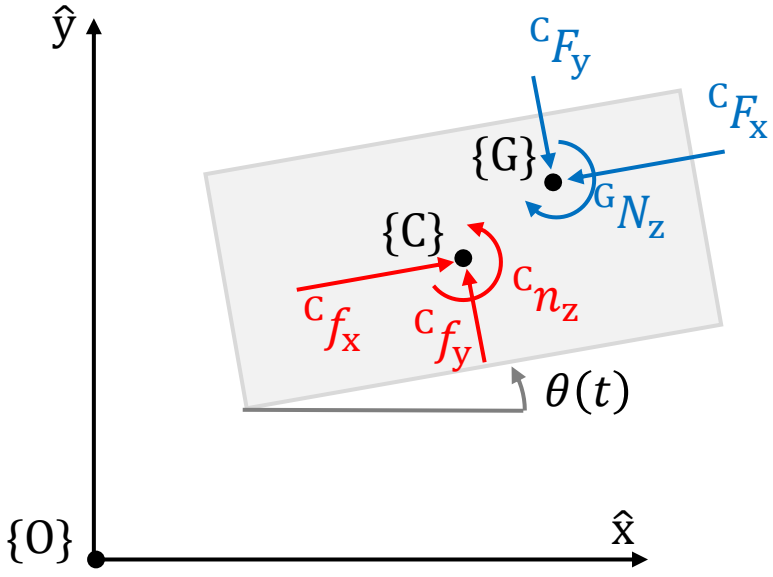
Write down the expressions for inertial force and torque

If the robot is accelerating,
then ${}^{C}F_{G}$ and ${}^{C}N_{G}$ are the inertial force and torque on the centre of mass {G}



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Write down the expressions for inertial force and torque



If the robot is accelerating,
then ${}^{C}F_{G}$ and ${}^{C}N_{G}$ are the inertial force and torque on the centre of mass {G}

$$ {}^{C}F_{G} = m\,{}^{C}\dot{v}_{G} \quad \begin{cases} {}^{C}F_{x} = m\,{}^{C}\ddot{x}_{G} \\ {}^{C}F_{y} = m\,{}^{C}\ddot{y}_{G} \end{cases} $$

$$ {}^{C}N_{G} = {}^{G}I\dot{\omega} + \omega \times {}^{G}I\omega \quad \Rightarrow \quad {}^{C}N_{z} = {}^{G}I\ddot{\theta} $$

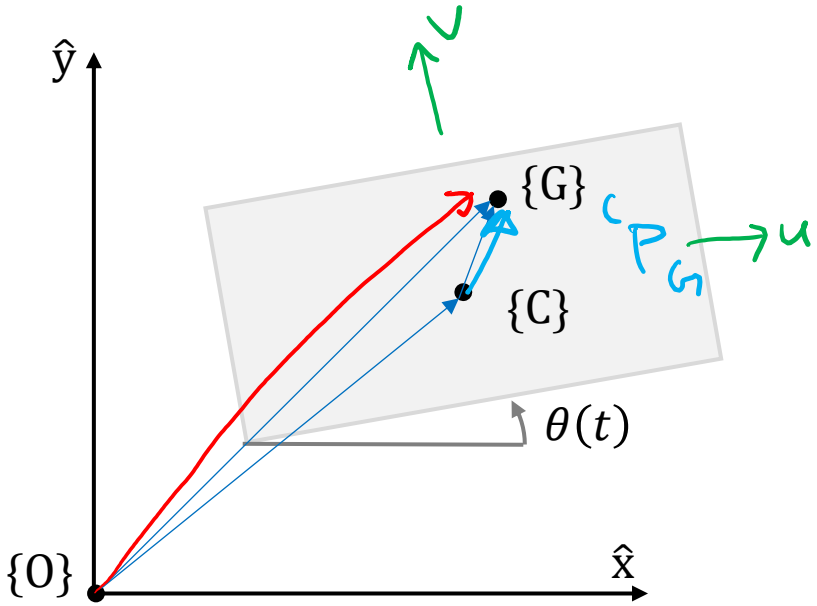
we know m, ${}^{G}I$
we don't know $\ddot{x}$, $\ddot{y}$, $\ddot{\theta}$

Given the robot's geometry and mass distribution, then we know m and ${}^{G}I$.
What we need to determine is the acceleration ${}^{C}\ddot{x}_{G}$ and ${}^{C}\ddot{y}_{G}$.

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Determining accelerations – first find velocity

Let us start by writing the position of the point {G} in frame {0}



$${}^0P_G = {}^0P_C + {}^0P_{C\text{toG}}$$

find derivative wrt time

$${}^0\dot{P}_G = {}^0\dot{P}_C + {}^0\dot{R}_C\,{}^CP_G + {}^0R_C\,{}^C\dot{P}_G$$

$${}^0\dot{P}_G = {}^0\dot{P}_C + {}^0\omega_C \times {}^0R_C\,{}^CP_G$$

$$\begin{Bmatrix} \dot{x} \\ \dot{y} \end{Bmatrix} \xrightarrow{J} \begin{Bmatrix} u \\ v \end{Bmatrix} = {}^C\dot{P}_C$$

$${}^0\dot{P}_G = J(\theta)\,{}^C\dot{P}_C + {}^0\omega_C \times {}^0R_C\,{}^CP_G$$

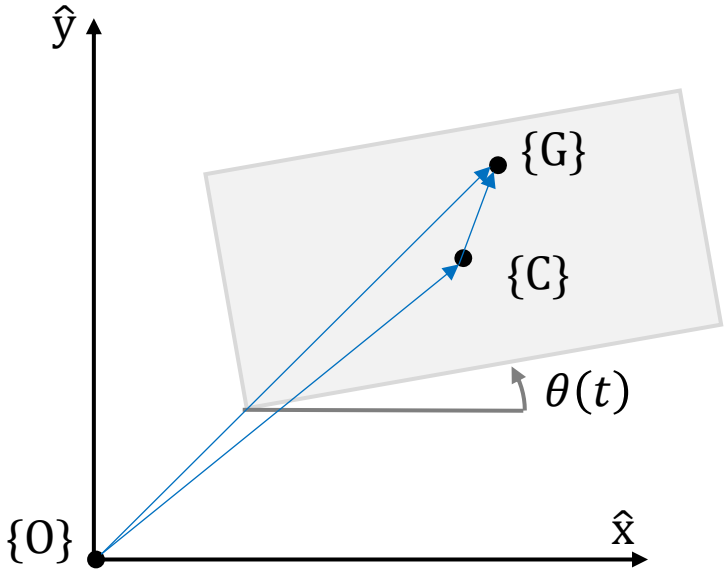
$$= J(\theta)\,{}^C\dot{P}_C + {}^0R_C\,{}^CR_0\,{}^0\omega_C \times {}^0R_C\,{}^CP_G$$

$$= {}^0R_C\,{}^C\dot{P}_C + {}^0R_C\,{}^C\omega_C \times {}^CP_G$$

$$= {}^0R_C\left[{}^C\dot{P}_C + {}^C\omega_C \times {}^CP_G\right]$$

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Determining accelerations – first find velocity



Let us start by writing the position of the point {G} in frame {O}

$$^{O}P_{G} = {}^{O}P_{C} + {}^{O}P_{G}$$

$$^{O}P_{G} = {}^{O}P_{C} + {}^{O}R_{C}\,{}^{C}P_{G}$$

Now, let us find the derivative w.r.t. time

$$^{O}\dot{P}_{G} = {}^{O}\dot{P}_{C} + {}^{O}\dot{R}_{C}\,{}^{C}P_{CG} + {}^{O}R_{C}\,{}^{C}\dot{P}_{G}$$

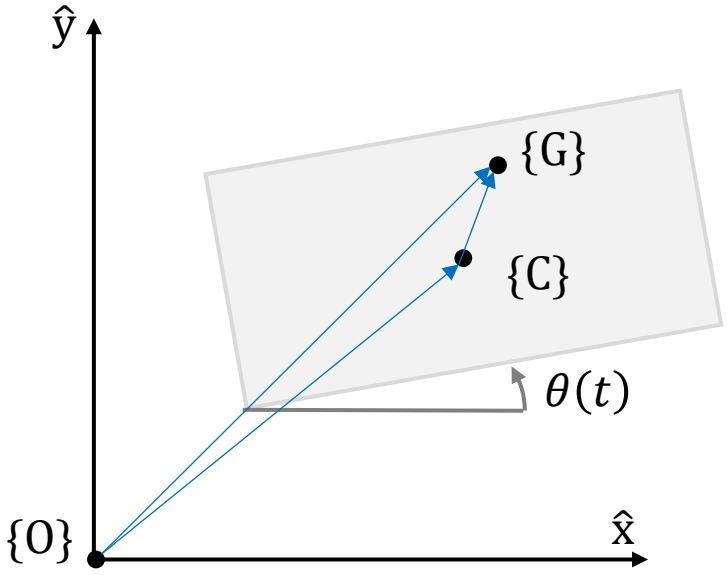
0

$$^{O}\dot{P}_{G} = {}^{O}\dot{P}_{C} + {}^{O}\dot{R}_{C}\,{}^{C}P_{G}$$

$$^{O}\dot{P}_{G} = {}^{O}\dot{P}_{C} + {}^{O}\omega_{C} \times {}^{O}R_{C}\,{}^{C}P_{G}$$

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Determining accelerations – first find velocity



$$ {}^{O}\dot{P}_{G} = {}^{O}\dot{P}_{C} + {}^{O}\omega_{C} \times {}^{O}R_{C}\,{}^{C}P_{G} $$

$$ {}^{O}\dot{P}_{G} = \mathbf{J}(\theta)\,{}^{C}\dot{P}_{C} + {}^{O}R_{C}\,{}^{C}R_{O}\,{}^{O}\omega_{C} \times {}^{O}R_{C}\,{}^{C}P_{G} $$

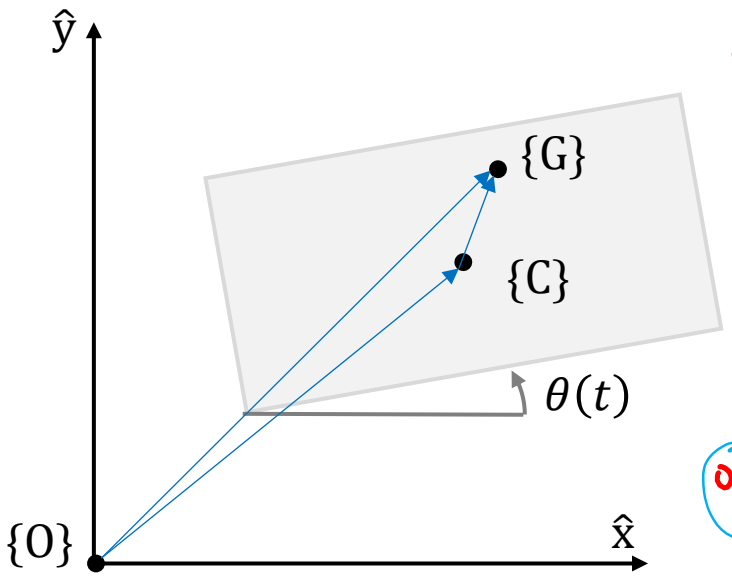
$$ {}^{O}\dot{P}_{G} = {}^{O}R_{C}\,{}^{C}\dot{P}_{C} + {}^{O}R_{C}\,{}^{C}\omega_{C} \times {}^{O}R_{C}\,{}^{C}P_{G} $$

$$ {}^{O}\dot{P}_{G} = {}^{O}R_{C}\left[{}^{C}\dot{P}_{C} + {}^{C}\omega_{C} \times {}^{C}P_{G}\right] $$

Let us differentiate again to get acceleration

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Differentiate velocity to obtain acceleration



$$ {}^{O}\dot{P}_G = {}^{O}R_C \left[{}^{C}\dot{P}_C + {}^{C}\omega_C \times {}^{C}P_G \right] $$

take the derivative wrt time

$$ {}^{O}\ddot{P}_G = {}^{O}\dot{R}_C \left[{}^{C}\dot{P}_C + {}^{C}\omega_C \times {}^{C}P_G \right] + {}^{O}R_C \left[{}^{C}\ddot{P}_C + {}^{C}\dot{\omega}_C \times {}^{C}P_G + {}^{C}\omega_C \times {}^{C}\dot{P}_G \right] $$

$$ = {}^{O}\dot{R}_C \left[{}^{C}\dot{P}_C + {}^{C}\omega_C \times {}^{C}P_G \right] + {}^{O}R_C \left[{}^{C}\ddot{P}_C + {}^{C}\dot{\omega}_C \times {}^{C}P_G \right] $$

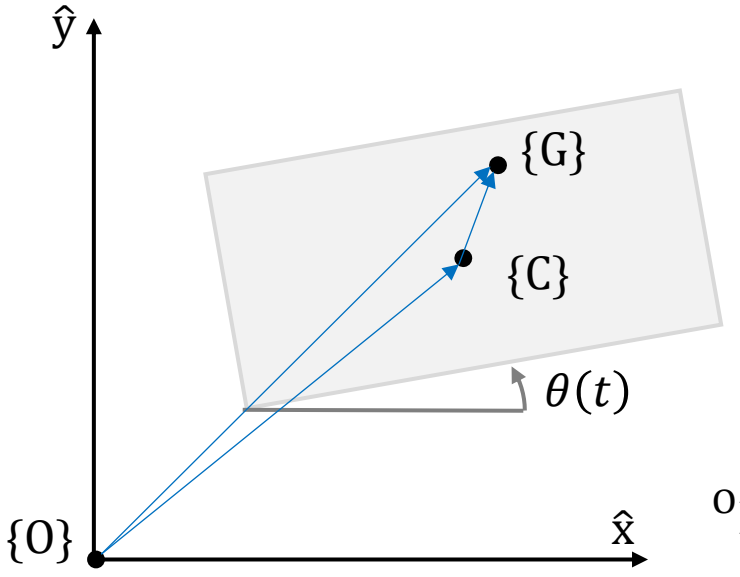
$$ {}^{O}\ddot{P}_G = \left[{}^{O}\omega_C \times {}^{O}R_C \right] \left[{}^{C}\dot{P}_C + {}^{C}\omega_C \times {}^{C}P_G \right] + {}^{O}R_C \left[{}^{C}\ddot{P}_C + {}^{C}\dot{\omega}_C \times {}^{C}P_G \right] $$

$$ {}^{O}R_C \, {}^{C}\ddot{P}_G = {}^{O}R_C \, {}^{C}\omega_C \times {}^{O}R_C \left[{}^{C}\dot{P}_C + {}^{C}\omega_C \times {}^{C}P_G \right] + {}^{O}R_C \left[{}^{C}\ddot{P}_C + {}^{C}\dot{\omega}_C \times {}^{C}P_G \right] $$

$$ {}^{C}\ddot{P}_G = {}^{C}\ddot{P}_C + {}^{C}\dot{\omega}_C \times {}^{C}P_G + {}^{C}\omega_C \times {}^{C}\dot{P}_C + {}^{C}\omega_C \times {}^{C}\omega_C \times {}^{C}P_G $$

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Differentiate velocity to obtain acceleration



$$ {}^{O}\dot{P}_G = {}^{O}R_C\left[{}^{C}\dot{P}_C + {}^{C}\omega_C \times {}^{C}P_G \right] $$

Take the derivative with respect to time

$$ {}^{O}\ddot{P}_G = {}^{O}\dot{R}_C\left[{}^{C}\dot{P}_C + {}^{C}\omega_C \times {}^{C}P_{CG} \right] + {}^{O}R_C\left[{}^{C}\ddot{P}_C + {}^{C}\dot{\omega}_C \times {}^{C}P_G + {}^{C}\omega_C \times {}^{C}\dot{P}_G \right] $$

0

$$ {}^{O}\ddot{P}_G = {}^{O}\dot{R}_C\left[{}^{C}\dot{P}_C + {}^{C}\omega_C \times {}^{C}P_{CG} \right] + {}^{O}R_C\left[{}^{C}\ddot{P}_C + {}^{C}\dot{\omega}_C \times {}^{C}P_G \right] $$

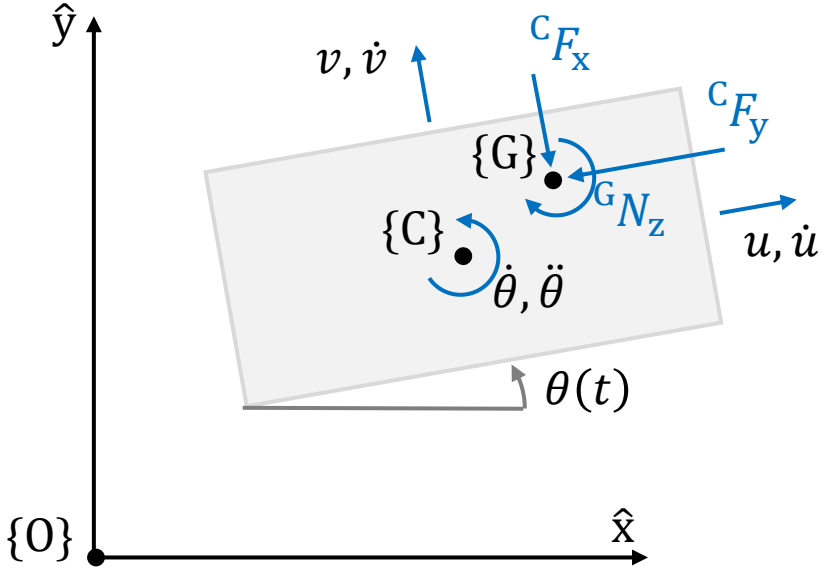
$$ {}^{O}\ddot{P}_G = {}^{O}\omega_C \times {}^{O}R_C\left[{}^{C}\dot{P}_C + {}^{C}\omega_C \times {}^{C}P_{CG} \right] + {}^{O}R_C\left[{}^{C}\ddot{P}_C + {}^{C}\dot{\omega}_C \times {}^{C}P_G \right] $$

$$ {}^{O}R_C\,{}^{C}\ddot{P}_G = {}^{O}R_C\,{}^{C}\omega_C \times {}^{O}R_C\left[{}^{C}\dot{P}_C + {}^{C}\omega_C \times {}^{C}P_{CG} \right] + {}^{O}R_C\left[{}^{C}\ddot{P}_C + {}^{C}\dot{\omega}_C \times {}^{C}P_G \right] $$

$$ {}^{C}\ddot{P}_G = {}^{C}\ddot{P}_C + {}^{C}\dot{\omega}_C \times {}^{C}P_{CG} + {}^{C}\omega_C \times {}^{C}\dot{P}_C + {}^{C}\omega_C \times {}^{C}\omega_C \times {}^{C}P_G $$

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Let us write down individual components



$$ {}^{C}\ddot{P}_{CG} = {}^{C}\ddot{P}_{C} + {}^{C}\dot{\omega}_{C} \times {}^{C}P_{CG} + {}^{C}\omega_{C} \times {}^{C}\dot{P}_{C} + {}^{C}\omega_{C} \times {}^{C}\omega_{C} \times {}^{C}P_{CG} $$

$$ \begin{bmatrix} {}^{c}\ddot{x}_{G} \\ {}^{c}\ddot{y}_{G} \\ 0 \end{bmatrix} = \begin{bmatrix} \dot{u} \\ \dot{v} \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \ddot{\theta} \end{bmatrix} \times \begin{bmatrix} {}^{c}x_{G} \\ {}^{c}y_{G} \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta} \end{bmatrix} \times \begin{bmatrix} u \\ v \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta} \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ \dot{\theta} \end{bmatrix} \times \begin{bmatrix} {}^{c}x_{G} \\ {}^{c}y_{G} \\ 0 \end{bmatrix} $$

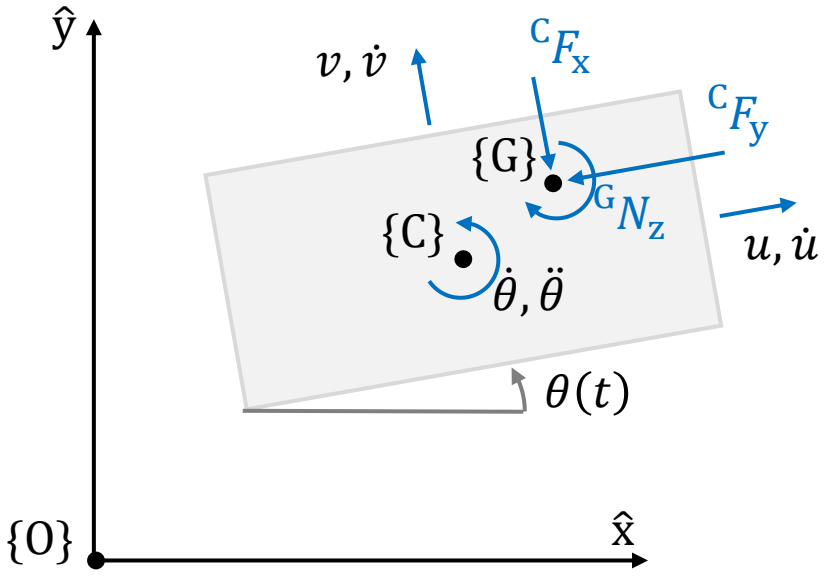
$$ {}^{c}\ddot{x}_{G} = \dot{u} - {}^{c}y_{G}\ddot{\theta} - v\dot{\theta} - {}^{c}x_{G}\dot{\theta}^{2} $$

$$ {}^{c}\ddot{y}_{G} = \dot{v} + {}^{c}x_{G}\ddot{\theta} + v\dot{\theta} - {}^{c}y_{G}\dot{\theta}^{2} $$

$$ {}^{c}F_{x} = m\,{}^{c}\ddot{x}_{G} \quad , \quad {}^{c}F_{y} = m\,{}^{c}\ddot{y}_{G} $$

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Let us write down individual components



$$ {}^{C}\ddot{P}_{G} = {}^{C}\ddot{P}_{C} + {}^{C}\dot{\omega}_{C} \times {}^{C}P_{G} + {}^{C}\omega_{C} \times {}^{C}\dot{P}_{C} + {}^{C}\omega_{C} \times {}^{C}\omega_{C} \times {}^{C}P_{G} $$

$$ \begin{bmatrix} {}^{C}\ddot{x}_{G} \\ {}^{C}\ddot{y}_{G} \\ 0 \end{bmatrix} = \begin{bmatrix} \dot{u} \\ \dot{v} \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \ddot{\theta} \end{bmatrix} \times \begin{bmatrix} {}^{C}x_{G} \\ {}^{C}y_{G} \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta} \end{bmatrix} \times \begin{bmatrix} u \\ v \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta} \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ \dot{\theta} \end{bmatrix} \times \begin{bmatrix} {}^{C}x_{G} \\ {}^{C}y_{G} \\ 0 \end{bmatrix} $$

$$ {}^{C}\ddot{x}_{G} = \dot{u} - {}^{C}y_{G}\ddot{\theta} - v\dot{\theta} - {}^{C}x_{G}\dot{\theta}^{2} $$

$$ {}^{C}\ddot{y}_{G} = \dot{v} + {}^{C}x_{G}\ddot{\theta} + v\dot{\theta} - {}^{C}y_{G}\dot{\theta}^{2} $$

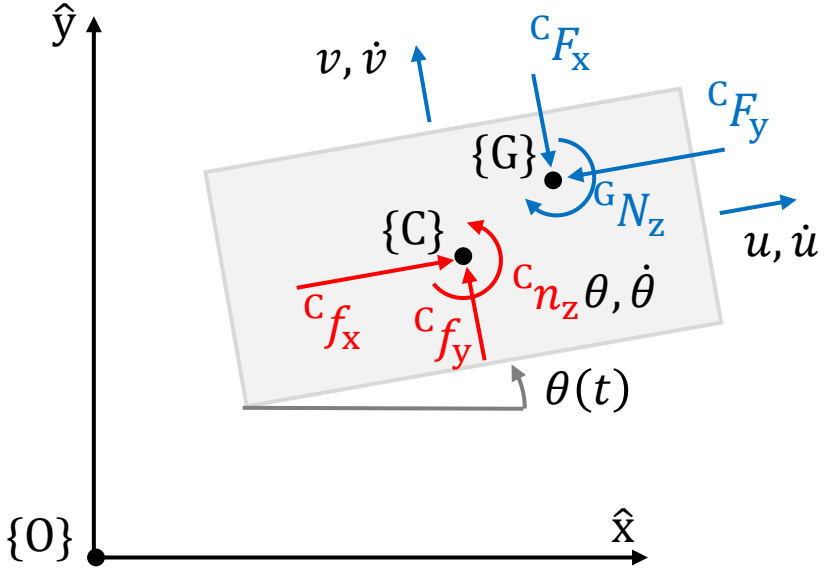
Therefore,

$$ {}^{C}F_{x} = m\,{}^{C}\ddot{x}_{G} = m\left(\dot{u} - {}^{C}y_{G}\ddot{\theta} - v\dot{\theta} - {}^{C}x_{G}\dot{\theta}^{2}\right) $$

$$ {}^{C}F_{y} = m\,{}^{C}\ddot{y}_{G} = m\left(\dot{v} + {}^{C}x_{G}\ddot{\theta} + v\dot{\theta} - {}^{C}y_{G}\dot{\theta}^{2}\right) $$

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Now let us write the equations of motion



$$
\begin{aligned}
{}^{\mathrm{C}}f_{\mathrm{x}} &= {}^{\mathrm{C}}F_{\mathrm{x}} = \mathrm{m}\left(\dot{u} - {}^{\mathrm{C}}\mathrm{y}_{\mathrm{G}}\ddot{\theta} - v\dot{\theta} - {}^{\mathrm{C}}\mathrm{x}_{\mathrm{G}}\dot{\theta}^2\right) \\
{}^{\mathrm{C}}f_{\mathrm{y}} &= {}^{\mathrm{C}}F_{\mathrm{y}} = \mathrm{m}\left(\dot{v} + {}^{\mathrm{C}}\mathrm{x}_{\mathrm{G}}\ddot{\theta} + v\dot{\theta} - {}^{\mathrm{C}}\mathrm{y}_{\mathrm{G}}\dot{\theta}^2\right) \\
{}^{\mathrm{C}}n_{\mathrm{z}} &= {}^{\mathrm{C}}N_{\mathrm{z}} - \left({}^{\mathrm{C}}\mathrm{y}_{\mathrm{G}}\,{}^{\mathrm{C}}f_{\mathrm{x}} - {}^{\mathrm{C}}\mathrm{x}_{\mathrm{G}}\,{}^{\mathrm{C}}f_{\mathrm{y}}\right) \\
&= {}^{\mathrm{G}}I\ddot{\theta} - \left({}^{\mathrm{C}}\mathrm{y}_{\mathrm{G}}\,{}^{\mathrm{C}}f_{\mathrm{x}} - {}^{\mathrm{C}}\mathrm{x}_{\mathrm{G}}\,{}^{\mathrm{C}}f_{\mathrm{y}}\right)
\end{aligned}
$$

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Equations of motion in matrix form

$$\begin{bmatrix} m & 0 & -m\,{}^{C}y_G \\ 0 & m & m\,{}^{C}x_G \\ -m\,{}^{C}y_G & m\,{}^{C}x_G & {}^{G}I + m\left({}^{C}x_G{}^{2} + {}^{C}y_G{}^{2}\right) \end{bmatrix} \begin{bmatrix} \dot{u} \\ \dot{v} \\ \ddot{\theta} \end{bmatrix} + \begin{bmatrix} -m\dot{\theta}\left(v + {}^{C}x_G\dot{\theta}\right) \\ m\dot{\theta}\left(u - {}^{C}y_G\dot{\theta}\right) \\ m\dot{\theta}\left({}^{C}x_G u + {}^{C}y_G v\right) \end{bmatrix} = \begin{bmatrix} {}^{C}f_x \\ {}^{C}f_y \\ {}^{C}n_z \end{bmatrix}$$

$$\mathbf{M}\dot{\boldsymbol{\zeta}} + c(\boldsymbol{\zeta}) = {}^{C}\mathbf{Q}$$

Depends only on geometry and mass distribution

Is a function of instantaneous velocity

Forces and moments w.r.t chassis frame

Note that all the terms here are written in the chassis frame {C}

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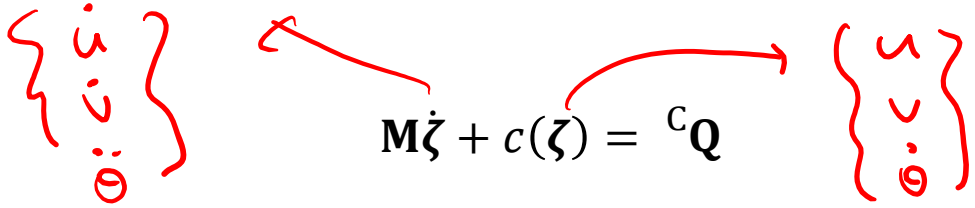
Homework

Use the Lagrangian formulation and verify if you arrive at the same set of equations.



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It is useful to express these quantities in the inertial frame {O}



Aim: to express the above expression in the form of ${}^{\mathrm{O}}\mathbf{Q} = function(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, \ddot{\boldsymbol{\xi}})$

To do that, we need

1. a way to represent ζ and $\dot{\zeta}$ in terms of $\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, \ddot{\boldsymbol{\xi}}$
2. a relation between ${}^{\mathrm{C}}\mathbf{Q}$ and ${}^{\mathrm{O}}\mathbf{Q}$

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Representing ζ and $\dot{\zeta}$ in terms of $\xi, \dot{\xi}, \ddot{\xi}$

$$\begin{Bmatrix} x \\ y \\ \theta \end{Bmatrix} \quad \dot{\xi} = J\zeta \quad \begin{Bmatrix} u \\ v \\ \dot{\theta} \end{Bmatrix}$$

$$\ddot{\xi} = \dot{J}\zeta + J\dot{\zeta}$$

$$J\dot{\zeta} = \ddot{\xi} - \dot{J}\zeta$$

$$\dot{\zeta} = J^{-1}[\ddot{\xi} - \dot{J}\zeta] = J^{-1}[\ddot{\xi} - \dot{J}J^{-1}\dot{\xi}]$$

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Representing ζ and $\dot{\zeta}$ in terms of $\xi, \dot{\xi}, \ddot{\xi}$

Previously (in lecture 10), we obtained the kinematic relation $\dot{\boldsymbol{\xi}} = \nu = \mathbf{J}(\boldsymbol{\xi})\zeta$, where $\boldsymbol{\xi} = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix}$ and $\zeta = \begin{bmatrix} u \\ v \\ \dot{\theta} \end{bmatrix}$

Taking the derivative of $\dot{\boldsymbol{\xi}} = \mathbf{J}(\eta)\zeta$ w.r.t time, we get

$$\ddot{\boldsymbol{\xi}} = \dot{\mathbf{J}}(\boldsymbol{\xi})\zeta + J(\boldsymbol{\xi})\dot{\zeta}$$

$$\dot{\zeta} = \mathbf{J}^{-1}(\boldsymbol{\xi})\left[\ddot{\boldsymbol{\xi}} - \dot{\mathbf{J}}(\boldsymbol{\xi})\zeta\right]$$

$$\dot{\zeta} = \mathbf{J}^{-1}(\boldsymbol{\xi})\left[\ddot{\boldsymbol{\xi}} - \dot{\mathbf{J}}(\boldsymbol{\xi})\mathbf{J}^{-1}(\boldsymbol{\xi})\dot{\boldsymbol{\xi}}\right]$$



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Relation between ${}^{C}\underline{\underline{Q}}$ and ${}^{O}\underline{Q}$

Total power consumed $= {}^{O}\underline{Q} \cdot \dot{\xi} = {}^{C}\underline{Q} \cdot \zeta$

$${}^{O}Q^{T}\dot{\xi} = {}^{C}Q^{T}\zeta$$

$${}^{O}Q^{T}J\zeta = {}^{C}Q^{T}\zeta$$

$${}^{O}Q^{T}J = {}^{C}Q^{T}$$

$$J^{T}\,{}^{O}Q = {}^{C}Q$$

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Relation between ${}^{C}\mathbf{Q}$ and ${}^{O}\mathbf{Q}$

To obtain a relation between ${}^{C}\mathbf{Q}$ and ${}^{O}\mathbf{Q}$, we consider the conservation of energy (power, to be precise)

The power consumed by the robot should be the same, irrespective of the frame used to calculate it.

$$ {}^{O}\mathbf{Q} \cdot \dot{\boldsymbol{\xi}} = {}^{C}\mathbf{Q} \cdot \zeta $$

$$ {}^{O}\mathbf{Q}^{\intercal} \dot{\boldsymbol{\xi}} = {}^{C}\mathbf{Q}^{\intercal} \zeta $$

$$ {}^{O}\mathbf{Q}^{\intercal} \mathbf{J}(\boldsymbol{\xi}) \zeta = {}^{C}\mathbf{Q}^{\intercal} \zeta $$

$$ {}^{O}\mathbf{Q}^{\intercal} \mathbf{J}(\boldsymbol{\xi}) = {}^{C}\mathbf{Q}^{\intercal} $$

$$ \mathbf{J}(\boldsymbol{\xi})^{\intercal} \, {}^{O}\mathbf{Q} = {}^{C}\mathbf{Q} $$

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Let us plug these into the equation of motion



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Let us plug these into the equation of motion

$$M\dot{\zeta} + c(\zeta) = {}^{C}\mathbf{Q}$$

$$M\left[\mathbf{J}^{-1}(\boldsymbol{\xi})\left[\ddot{\boldsymbol{\xi}} - \dot{\mathbf{J}}(\boldsymbol{\xi})\mathbf{J}^{-1}(\boldsymbol{\xi})\dot{\boldsymbol{\xi}}\right]\right] + c\left(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}\right) = \mathbf{J}(\boldsymbol{\xi})^{\intercal}\,{}^{O}\mathbf{Q}$$

$$(\mathbf{J}(\boldsymbol{\xi})^{\intercal})^{-1}M\mathbf{J}^{-1}(\boldsymbol{\xi})\ddot{\boldsymbol{\xi}} + (\mathbf{J}(\boldsymbol{\xi})^{\intercal})^{-1}\left[c\left(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}\right) - M\dot{\mathbf{J}}(\boldsymbol{\xi})\mathbf{J}^{-1}(\boldsymbol{\xi})\dot{\boldsymbol{\xi}}\right] = {}^{O}\mathbf{Q}$$

$${}^{O}M\ddot{\boldsymbol{\xi}} + {}^{O}c\left(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}\right) = {}^{O}\mathbf{Q}$$

This is now entirely in terms of $\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, \ddot{\boldsymbol{\xi}}$

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How do we use these in practice?

It is common to consider the dynamics equation in the chassis frame.

This is because, the forces and moments on the robot are due to the interaction of the wheels with the ground.

Say, we start from a known configuration and that the forces from the wheels are given.

$$\dot{\zeta} = M^{-1}\left[{}^{C}\mathbf{Q} - c(\zeta) \right]$$

Then, integrate $\dot{\zeta}$ to obtain ζ. From this obtain $\dot{\boldsymbol{\xi}}$ using $\dot{\boldsymbol{\xi}} = J(\boldsymbol{\xi})\zeta$.

Finally integrate $\dot{\boldsymbol{\xi}}$ to obtain the new position of the robot $\boldsymbol{\xi}$.



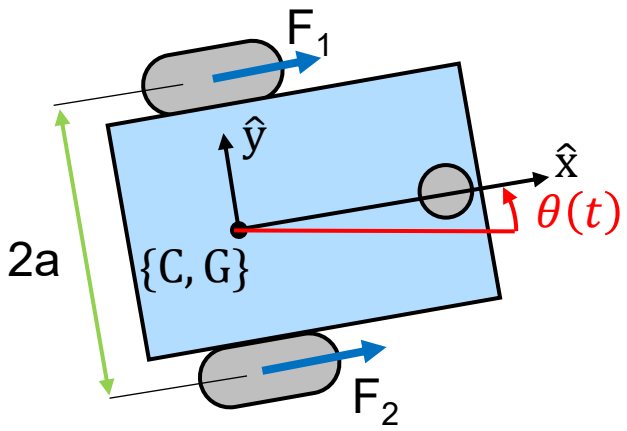
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Where is the torque on the chassis coming from?



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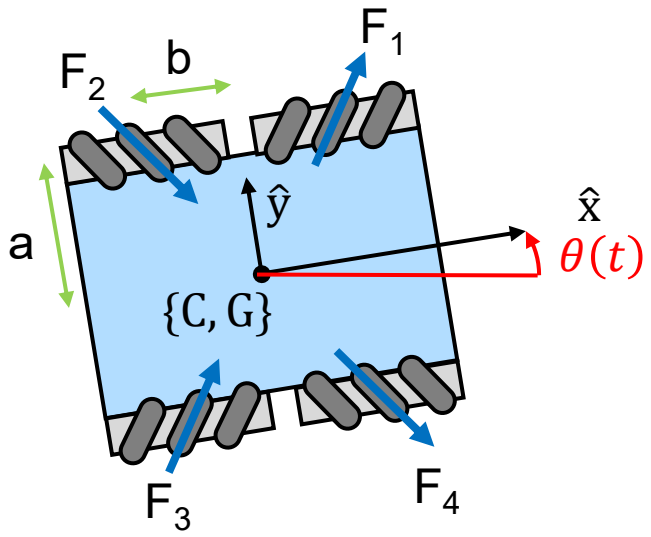
Example: Differential drive robot



$$\begin{Bmatrix} {}^{c}f_x \\ {}^{c}f_y \\ n_z \end{Bmatrix}_{3\times 1} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ -a & a \end{bmatrix}_{3\times 2} \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}_{2\times 1}$$

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Example: Swedish wheel robot



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Summary

- Dynamics equation in the body frame
- Dynamics equation in the inertial frame



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Next lecture

- Planning algorithms



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